

The Innocence Project
Response to Request for Information on the
Development of an Artificial Intelligence (AI) Action Plan
Submitted March 14, 2025

The Innocence Project (IP) is pleased to respond to the Request for Input (RFI) from the Office of Science and Technology Policy (OSTP) regarding the development of an Artificial Intelligence (AI) Action Plan.¹ As indicated in the RFI, the Trump Administration has set a goal to establish government policies so that “the United States can solidify its position as the leader in AI and secure a brighter future for all Americans.” It is possible for the United States to lead the AI industry while still ensuring safe and reliable practices and managing risks, especially when developing or procuring AI systems that impact the liberty, freedoms, and safety of Americans.

We believe, and our experience has shown, that careful risk assessment and establishing appropriate guardrails around the deployment of new AI technologies are crucial to developing AI tools that will lead to these measures of success. Nowhere is this more important than in the criminal justice system, where they are necessary for ensuring public safety and preventing wrongful convictions.

For over 30 years, IP has worked to exonerate innocent individuals and prevent false convictions through institutional reform. In cases where we have conclusively proven innocence, the misapplication of forensic science has contributed to 52% of the unjust underlying convictions.² Because unreliable and flawed forensic methods that are not grounded in science have historically contributed significantly to wrongful convictions and other miscarriages of justice, we are deeply concerned about the increasing use of AI technologies in the justice system that have not been subject to robust evaluation and validation, intentional oversight, and rigorous regulation. We have already seen alarming instances of advanced AI technologies, such as facial recognition systems, being misused and leading to false arrests.

¹ This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

² Innocence Project, Overturning Wrongful Convictions Involving Misapplied Forensics, Innocence Project, <https://innocenceproject.org/misapplication-of-forensic-science/>.

The Innocence Project firmly believes that the development and implementation of AI-powered applications in the criminal legal system must prioritize validity and reliability; be grounded in a solid scientific foundation; and consider ethical, legal, and social implications (ELSI). All of these will be crucial to ensuring the accuracy and excellence necessary to meet the AI Action Plan's goal to "promote human flourishing, economic competitiveness, and national security."

Recommendations

Below we summarize our recommendations in several of the key areas identified in the RFI:

1. **Explainability and Accuracy of AI Model Outputs** – AI models should be developed with transparent methodologies that are explainable and have undergone independent validation and necessary auditing to assess their accuracy and reliability. Doing so will foster trust and accountability in forensic science applications.
2. **Data Privacy and Security** – AI systems should adhere to strict privacy safeguards throughout their development and deployment, including ensuring data minimization, implementing robust encryption protocols, and restricting access to authorized personnel. A risk-based approach to privacy impact assessments should be mandated to prevent unauthorized data exposure or misuse, especially in cases involving minors, vulnerable populations, and sensitive criminal investigations.
3. **Regulation and Governance** - The federal government should establish a reliable and usable framework that requires forensic AI developers to undergo rigorous testing, certification, and compliance assessments before law enforcement agencies and forensic laboratories adopt their tools.
4. **Technical and Safety Standards** - Standardized benchmarks must be established to evaluate the technical robustness and safety of forensic AI systems, including the development of industry-wide guidelines focused on data quality, bias mitigation, and model interpretability.
5. **Research and Development** – In partnership with academia, law enforcement, and independent researchers, the federal government should fund research initiatives to develop innovative AI techniques for forensic analysis while maintaining transparency, reducing algorithmic bias, and addressing ethical concerns.
6. **Procurement** – The federal procurement process should prioritize AI solutions that meet high ethical, legal, and technical standards. Vendors should be required to report their training data, model performance, bias mitigation strategies, and compliance with evolving forensic standards.

These focus areas are critical for advancing the responsible integration of AI in forensic science, addressing ethical concerns, and mitigating potential harms to public safety, such as incorrect suspect identification which leads to the actual perpetrator remaining undetected.

1. Explainability and Accuracy of AI Model Outputs

AI systems used in criminal investigations, evidence analysis, and judicial processes are opaque. Due to their proprietary nature, judges, defendants, and lawyers know little about how AI tools operate. As a result, judges have difficulty deciphering and questioning algorithmic results and inadvertently end up permitting private actors to influence sentencing outcomes without traditional accountability mechanisms.³

Another major obstacle to explainability is the black-box nature of these AI systems. The AI black box problem refers to the inability to see how some AI systems make their decisions. Many machine learning algorithms are indecipherable, particularly popular deep learning neural network methods.⁴ The processes that occur in these boxes are self-driven, and most programmers, data scientists, and users cannot interpret them. The lack of explainability makes it difficult to assess inputs adequately and to determine from where potential problems stem. Black box models that preclude a clear understanding of their decision-making processes warrant additional front-end and ongoing scrutiny.

Developers of AI models need to construct methodologies that are accessible to forensic experts, legal practitioners, and jurors, allowing them to understand the reasoning behind the models' conclusions. *Transparency* and *explainability* of AI systems are two critical pillars of a functioning AI system. The third is *accuracy and reliability*, which requires implementing mechanisms that allow for independent validation and ongoing auditing of their AI model outputs.

Studies of AI tools have raised serious concerns about their reliability and accuracy. For example, research has shown that the recidivism and pretrial failure accuracy rates predicted by popular tools such as the Pre-Trial Risk Assessment Instrument (PRAI) and Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) are no better than those achieved by human predictions.⁵ A 2022 systematic review of validation studies on 11 commonly used sentencing risk assessment tools concluded that the

³ Andrea Nishi, *Privatizing Sentencing: A Delegation Framework for Recidivism Risk Assessment*, 119 SSRN JOURNAL (2019), <https://www.ssrn.com/abstract=3335946>.

⁴ Arun Rai, *Explainable AI: From Black Box to Glass Box*, 48 J. ACAD. MARK. SCI. 137 (2020).

⁵ Practitioner's Guide to COMPAS Core, (2019).

predictive performance of these tools varied significantly, with results ranging from poor to moderate.⁶

All algorithms that depend on probabilistic calculations or inferences carry inherent uncertainties that affect their accuracy. If one does not grasp the probabilistic nature of predictive algorithms, it may lead to overinterpreting or overreliance on the data and predictions. Additionally, this misunderstanding can result in not taking the necessary steps to fully understand, verify, or confirm those predictions.

Predictive algorithms pose significant risks, especially when users underestimate or overlook their inherent probabilistic nature. This misunderstanding can create a false sense of certainty, leading individuals to believe that predictions are definitive. Such misconceptions can result in wrongful arrests and harm innocent individuals. It is crucial to approach these technologies with caution and to understand their limitations. Regardless of the accuracy of the coding, using systematically biased data can lead to skewed or incorrect conclusions, perpetuating false narratives about the criminality of entire communities.⁷

Lastly, within the past few years, the forensic community has been fixated on artificial intelligence. Some see it as a way to resolve issues relating to human error and bias that have previously led to miscarriages of justice. Others see it as a way to streamline and automate tasks. However, using AI to enhance an unscientific and flawed forensic technique will not ensure or improve reliability, especially if the same or similar methods and principles are applied. AI cannot be utilized to conduct analyses based on invalid or unproven claims.

2. Data Privacy and Security Throughout AI System Development

Forensic science often involves sensitive personal data, including biometric information, crime scene evidence, and DNA profiles. It is essential for AI systems to adhere to strict privacy safeguards throughout their development and deployment. This includes ensuring data minimization, implementing robust encryption protocols, and restricting access to authorized personnel. A risk-based approach to privacy impact assessments should be mandated to prevent unauthorized data exposure or misuse, especially in cases involving minors, vulnerable populations, and sensitive criminal investigations.

⁶ Seena Fazel et al., The Predictive Performance of Criminal Risk Assessment Tools Used at Sentencing: Systematic Review of Validation Studies, 81 JOURNAL OF CRIMINAL JUSTICE 101902 (2022).

⁷ Grace Baek & Taylor Mooney, LAPD Not Giving up on Data-Driven Policing, Even after Scrapping Controversial Program, (2020), <https://www.cbsnews.com/news/los-angeles-police-department-laser-data-driven-policing-racial-profiling-2-0-cbsn-originals-documentary/>.

Additionally, the unregulated access of law enforcement to commercial databases containing personal biometric information raises significant concerns regarding due process and Fourth Amendment protections. Without judicial oversight, law enforcement agencies can exploit these databases to conduct searches without proper legal justification, which increases the risk of false identifications and privacy violations. The government should establish strict regulations ensuring that law enforcement's use of AI-driven forensic databases is subject to warrants, transparency requirements, and independent audits. Furthermore, commercial data brokers should be required to implement safeguards that prevent the misuse of personal data by unauthorized entities.

Several examples highlight how law enforcement's use of commercial databases has raised constitutional concerns:

- Clearview AI and Warrantless Facial Recognition Searches: Clearview AI, a facial recognition company, collected billions of images from the internet and sold access to them for use by law enforcement agencies. The absence of warrants and oversight in the use of this technology has raised concerns about mass surveillance and potential violations of Fourth Amendment protections against unreasonable searches.⁸
- CLEAR System and ICE Surveillance: ICE has used the CLEAR database from Thomson Reuters to track individuals, including undocumented immigrants, without obtaining warrants. The reliance on aggregated commercial data, often without consent, has been criticized as an unconstitutional invasion of privacy.^{9 10}
- AI-Driven Predictive Policing: AI-driven predictive policing tools, such as PredPol, utilize historical crime data to forecast potential future crime hotspots. However, experts have raised concerns regarding these systems, noting concerns about their inaccuracy and disproportionate focus on communities of color.¹¹ This can result in wasteful and ineffective over-policing and raises important issues related to unreasonable searches and systemic bias.¹²

3. Regulation and Governance

To maintain public trust and ensure accuracy that will lead to economic competitiveness and strong national security, AI tools used in forensic science must be governed by clear,

⁸ Kashmir Hill, Facial Recognition Start-Up Mounts a First Amendment Defense, The New York Times (2020), <https://www.nytimes.com/2020/08/11/technology/clearview-floyd-abrams.html>.

⁹ Sidney Fussell, A Border Town Confronts the Reality of Police Surveillance, WIRED (2021), <https://www.wired.com/story/border-town-confronts-reality-police-surveillance/>.

¹⁰ Joseph Cox, 'Fourth Amendment Is Not For Sale Act' Would Ban Clearview and Warrantless Location Data Purchases, Vice News (2021), <https://www.vice.com/en/article/fourth-amendment-is-not-for-sale-act-would-ban-clearview-and-warrantless-location-data-purchases/>.

¹¹ H Aaron Sankin & Surya Mattu, Predictive Policing Software Terrible at Predicting Crimes, Wired (2023), <https://www.wired.com/story/plainfield-geolitea-crime-predictions/>.

¹² Tim Lau, Predictive Policing Explained, Brennan Center for Justice (2020), <https://www.brennancenter.org/our-work/research-reports/predictive-policing-explained>.

enforceable regulations. The federal government should establish a reliable and usable framework that requires forensic AI developers to undergo rigorous testing, certification, and compliance assessments before law enforcement agencies and forensic laboratories adopt their tools. Further, AI decision-making processes should be subject to independent oversight, ensuring they align with constitutional protections and due process rights.

More specifically, a national oversight system for forensic science should include an accountability framework to address errors, issues, or instances of negligence or misconduct. It is the ethical and professional obligation of all stakeholders in the criminal legal system to correct mistakes and provide notifications when necessary. The criminal legal system should not be left to its own devices in holding itself accountable; instead, an external oversight system should be established. This could involve the establishment of an AI advisory committee with specific focus on implementation in law enforcement settings. In doing this, the administration should ensure that the entity has an independent voice and the ability to address urgent constitutional issues. Under the previous administration, an initial report was released that left the examination of implementation of AI technologies in the criminal justice system incomplete; this crucial step should now be completed.¹³

4. Technical and Safety Standards

Standardized benchmarks must be established to evaluate the technical robustness and safety of forensic AI systems. This includes the development of industry-wide guidelines focused on data quality, bias mitigation, and model interpretability. AI systems should undergo adversarial testing to identify vulnerabilities and potential avenues for misuse. Given their extensive experience in working with industries and stakeholders, the National Institute of Standards and Technology (NIST) should lead efforts to determine the best scientific and technical approaches for implementing large models in federal agencies, particularly regarding how emerging technologies can be used by law enforcement.

5. Research and Development

Investment in research is essential to advancing the responsible use of AI in forensic science and to solidifying the United States' leadership position in AI. The federal government should fund interdisciplinary research initiatives that explore novel AI techniques for forensic analysis while addressing ethical concerns. Research should focus on improving forensic AI transparency, reducing algorithmic bias, and developing methods

¹³ Miriam Vogel, et al, NAIAC Law Enforcement Subcommittee: Year 1 Report and Roadmap, (2024), <https://ai.gov/wp-content/images/NAIAC-LE-Subcommittee-Year1-Report-Roadmap.pdf>.

to detect AI-generated deepfakes that could compromise the integrity of evidence. Partnerships between academic institutions, law enforcement, independent researchers, and the private sector should be encouraged to drive innovation while upholding forensic best practices.

6. Procurement

The federal procurement process should prioritize AI solutions that meet high ethical, legal, and technical standards. Agencies procuring AI tools for forensic science should require vendors to disclose detailed information about their training data, model performance, and bias mitigation strategies to defense counsel. Further, procurement contracts should include ongoing monitoring requirements to ensure AI systems maintain compliance with evolving forensic standards. To promote accountability, AI vendors should be required to submit impact assessments to the procuring entity outlining how their technologies affect forensic accuracy, public safety, and civil liberties.

Relying on AI systems for sentencing, pretrial, and posttrial decisions raises significant concerns, especially when their results have not been independently validated. It is essential to conduct independent validation studies before these systems are implemented. To facilitate this, some states have established "validation committees" responsible for determining the appropriate usage of these tools, assessing how to weigh predicted risk scores, and evaluating their accuracy rates.¹⁴

Validation should occur before procurement, or at the very least, before deployment, and the results must be made publicly available. These efforts will lead to a better understanding of the accuracy and reliability of these tools. Knowing the accuracy and error rates of these systems can help prevent judges from falling victim to confirmation biases.

Conclusion

As noted in the RFI, "The Trump Administration recognizes that with the right government policies, the United States can solidify its position as the leader in AI and secure a brighter future for all Americans." Advancements in technology have always brought both opportunities and risks, and AI is no exception. It is one of today's most powerful technologies, with the potential to improve lives and address some of society's biggest

¹⁴ Brian Lovins & Lori Lovins, Riverside Pretrial Assistance to California Counties (PACC) Project: Validation of a Pretrial Risk Assessment Tool, (2016), https://www.cri.org/assets/2017/07/6_Riverside_Validation_Final_Report_5-3-16.pdf, Victoria Terranova & Kyle Ward, Colorado Pretrial Assessment Tool Validation Study Final Report, (2020), https://www.nacdl.org/getattachment/18510570-e0eb-4d40-b737-5aafb30c1085/terranoaward_cpat-validation-stud_y_final-report.pdf.

challenges. At the same time, the misuse of AI has the potential to increase threats to freedom, safety and security, infringe upon civil rights and privacy, erode public trust in the justice system, and weaken economic competitiveness.

The Innocence Project appreciates the Administration's commitment to consulting with a broad group of stakeholders as it seeks to advance America's leadership in the field of AI. We look forward to further engagement and collaboration on this matter to ensure forensic AI tools are reliable, make communities safer, and avoid wrongful convictions.